

Software Requirements Specification (SRS)

Project: Academic & Campus Event Management System

1. Introduction

CampusHub is a web-based platform that integrates academic management and campus event management into a single system for students, faculty, event coordinators, and administrators.

2. Purpose

To centralize academic activities, event management, notifications, and communication while reducing missed deadlines and manual work.

3. Scope

- Academic management
- Campus event management
- Role-based access control
- Assignment and study material management
- Attendance and marks management
- Automated reminders and notifications

4. Problem Statement

- Multiple communication channels cause confusion.
- Students miss assignment and event deadlines.
- Manual management consumes staff time.
- Academic and event information is scattered.

5. Proposed Solution

CampusHub provides a centralized platform with dashboards, notifications, assignment tracking, event registration, study materials, attendance, marks, and reports.

6. User Roles

Admin

- Manage departments, courses, faculty, students, reports

Faculty

- Assignments, attendance, marks, study materials, announcements

Event Coordinator

- Create events, registrations, event timeline, certificates

Student

- Assignments, submissions, event registration, notifications, attendance, marks

7. Functional Requirements

- Secure login and JWT authentication
- Role-based dashboards
- Assignment creation and submission
- Attendance management
- Marks management
- Study material repository
- Event creation and registration
- Event timeline tracking
- Email, browser, and in-app notifications
- Reports and analytics

8. Non-Functional Requirements

- Responsive user interface
- Secure authentication
- Scalable architecture
- Reliable database
- Fast response time
- Easy maintenance

9. External APIs

- **Public Holiday API:** Displays national/state holidays in academic calendar.
- **Google Calendar API:** Adds assignments and events to Google Calendar.
- **Cloudinary API:** Stores images, files, certificates, and study materials.
- **OpenStreetMap + Nominatim API:** Displays event venues and navigation.

10. Technology Stack

Layer	Technology
Frontend	React.js, CSS, Fetch API
Backend	Spring Boot, Spring Security, Spring Data JPA
Database	PostgreSQL
Authentication	JWT
Notifications	JavaMail, Spring Scheduler
Version Control	Git & GitHub

11. System Requirements

- Java 17
- Spring Boot 3.5.x
- PostgreSQL

- Node.js & npm
- Modern Web Browser
- Windows 10/11

12. Future Enhancements

- Mobile application
- Push notifications
- QR-based attendance
- AI-based academic assistant
- Placement management integration

13. Conclusion

CampusHub provides a unified platform for academic and campus event management, improving communication, reducing missed deadlines, and enhancing the overall campus experience.

14. Overall System Workflow



